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Roll No

CE-6004-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Highway Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable missing values.

1. A national highway passing through a flat terrain has a horizontal curve of radius equal to the ruling minimum radius. If the design speed is 100 kmph. Calculate:

- a) Minimum sight distance
- b) Superelevation
- c) Extra widening
- d) Length of transition curve (Assume rate of introduction of S.E. as 1 in 150 and rotation of pavement is around to the mid section). 14

2. a) What do you mean by WMM? Discuss its construction steps. 7

- b) Differentiate flexible and rigid pavement. 7

3. What do you mean by soil stabilization? Explain different methods of soil stabilization in details. 14

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4. a) What are the factors affecting runway alignments?
Explain. 7
- b) Explain basic runway length and its corrections. 7
5. What are the important of traffic control on airport? Explain
different types of traffic control devices used on airport. 14
6. Explain the followings: 14
 - a) Seal coat
 - b) Tack coat
 - c) Prime coat
 - d) Wearing coats
7. a) What are the recent developments in cement concrete
pavement design? Discuss. 7
- b) What are different vehicular characteristics which affect
the road design? Briefly explain. 7
8. Discuss requirements and importance of highway drainage. 14
